

miniMorse — Project Summary v1

Overview

miniMorse is a battery-powered ESP32 project where multiple standalone devices communicate wirelessly using Morse code in a shared chat room. The project is cloud-free and router-free, focusing on low-level IoT communication.

Platform and Technology Choices

Programming environment: Arduino (ESP32 core)
Wireless technology: ESP-NOW
Communication model: Broadcast
Network: No Wi-Fi network, no router, no IP
ESP-IDF: Not used in version 1

Core Communication Model

All devices send and receive ESP-NOW packets via broadcast.
All packets are heard by all devices within range.
Each device filters which packets are relevant.
Communication is connectionless and best effort.

Rooms and Ownership

Version 1: Owner = room.
A room is identified by the device that created it.
One room per owner.
When the owner leaves, the room ceases to exist.
No ownership transfer in v1.

Joining

Only one way to join.
No connection + short button press creates or joins a room.
Joining is sequential.

Leaving a Room

Long button press (~5 s).
Long inactivity timeout.

Chat Room Behavior

Multiple devices can participate.
All participants may write simultaneously.
First come, first serve.
All messages are public.

Morse and Local Interaction

Button acts as a Morse key.
Local decoding.
Immediate local display.
Network used only for completed messages.

Message Completion and READY State

Message defined by long pause.
READY state entered after pause.
Short press sends message.
Long press deletes message.

Timeout Behavior

Timeout while typing or in READY state leaves the room.

UI and State Flow

IDLE → CREATE/JOIN → IN_ROOM → READY → SEND/DELETE → IN_ROOM.
Timeout from IN_ROOM or READY → IDLE.

Version 1 Constraints

One room at a time.
One join method.
No secure pairing.
No ownership transfer.
No private messages.
No editing.
No buffering or retransmission.

Purpose and Learning Goals

Understand connectionless communication.
Learn ESP-NOW.
Build custom protocol and state logic.
Separate UX, logic, and radio.
Create a simple but consistent system.